



Environmental modulation of cytotoxicity and virulence factor stability in psychrotolerant *Bacillus cereus* group isolates

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ARTICLE INFO

Keywords:

Bacillus cereus
Psychrotolerant
Cytotoxicity
Gene transcription
hblC
Enterotoxins

ABSTRACT

The *Bacillus cereus* group comprises strains relevant to food spoilage, agriculture, and human disease, with psychrotolerant members associated with refrigerated food spoilage. However, the food safety risk posed by these psychrotolerant strains remains poorly understood. This study characterized psychrotolerant *B. cereus* group strains from *panC* groups II and VI by quantifying (i) cytotoxicity at different temperatures and media (10, 32, and 37 °C in brain heart infusion broth (BHIB), and 10 °C and 32 °C in skim milk broth (SMB)), (ii) transcription of virulence genes at these same conditions, and (iii) reduction in supernatant cytotoxicity due to protein stressors. Isolates showed higher cytotoxicity ($p = 0.00086$) when grown in BHIB at 32 °C (0.40 ± 0.23) compared to 37 °C (0.28 ± 0.20). A subset of tested isolates also showed higher cytotoxicity ($p = 0.0054$) and transcription ($p < 0.001$) of tested virulence genes at 32 °C than at 10 °C in SMB and BHIB. Thermal (65, 75, 100 °C for 1–2 min), low-pH (2), and trypsin (10 µg/µL) treatments significantly reduced cytotoxicity of supernatants collected from isolates grown at 32 °C in BHIB, though residual cytotoxicity remained above the level of sterile BHIB. This study highlights that psychrotolerant *B. cereus* isolates express cytotoxic virulence factors, although at a low level, in a dairy food model under conditions mimicking temperature abuse during refrigerated storage. Furthermore, the persistence of residual cytotoxicity following thermal, pH, and proteolytic treatments highlights the need to better understand the biological relevance of these residual activities during human exposure.

1. Introduction

The *Bacillus cereus* group, also referred to as *B. cereus sensu lato* (*s.l.*), is a diverse group of spore-forming bacteria that are classified into 8 phylogenetic groups, defined based on their *panC* sequence (Carroll et al., 2022). Strains belonging to different phylogenetic groups vary in their ecological niches, virulence potential, and physiological characteristics (Carroll et al., 2022). Some pathogenic *B. cereus s.l.* strains produce cereulide, a heat-stable toxin that is pre-formed in food and causes emetic intoxication with symptoms of vomiting and nausea. In contrast, diarrheal toxicoinfection typically results from strains that, once ingested, secrete enterotoxins such as non-hemolytic enterotoxin (Nhe), hemolysin BL (Hbl), cytotoxin K (CytK), along with additional virulence factors like sphingomyelinase (SMase) and phosphatidylcholine-specific phospholipase C (PC-PLC), among others.

Enterotoxins disrupt epithelial cell membranes and activate the NLRP3 inflammasome, which leads to symptoms of diarrhea (Fox et al., 2020; Mathur et al., 2019). Unlike intoxication, enterotoxin-mediated illness is considered a toxicoinfection, requiring viable bacteria to survive gastrointestinal passage and actively produce toxins in the intestine (Dietrich et al., 2021).

Psychrotolerant strains of *B. cereus* are a concern to the food industry because their spore-forming ability enables survival during thermal processing, and their subsequent growth can lead to spoilage of refrigerated foods (Takahashi et al., 2020). Psychrotolerant strains are commonly found in *panC* phylogenetic groups II and VI, represented by species *B. wiedmannii* and *B. mycoides* (previously reported as *B. weihenstephanensis*), respectively, which are adapted to growth at low temperatures (Benoit et al., 2019; Guinebreteiere et al., 2008). Psychrotolerant isolates harboring virulence genes have been isolated from a wide

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